



Soongsil University
B.S. in AI Convergence

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🔗 GitHub Profile

EDUCATION

- **Soongsil University** Mar. 2022 – Present
B.S. in AI Convergence | GPA: 3.7/4.5 (Major 3.9/4.5) Seoul, South Korea

PROJECTS

- **Jeju Dialect → English Bridge Translation** Oct. 2025 – Dec. 2025
Team Project
– Fine-tuned KoBART, performed data collection & preprocessing, and applied Automated Prompt NMT techniques inspired by recent research.
- **Naver Movie Review Sentiment Analysis (Kaggle)** Oct. 2025 – Nov. 2025
Team Project
– Achieved 2nd place with 0.97933 using a 5-fold ensemble of RoBERTa, KoELECTRA, and electra-ko.
- **EMNIST Autoencoder-based Anomaly Detection Optimization** Oct. 2025 – Dec. 2025
Team Project
– Conducted hyperparameter tuning and architecture experiments for reconstruction and anomaly detection.
– Analyzed relationships between metrics (accuracy, precision, recall, F1, threshold) and model parameters.
- **Medication Support System for Dementia Patients (Raspberry Pi + TFLite)** Oct. 2025 – Nov. 2025
Team Project
– Developed two DQN agents for alert-stage scheduling and medication timing on-device.
– Defined full state–action–reward structures with target networks and epsilon-greedy.
– Implemented nightly training via cron using buffered user logs and deployed TFLite models on Raspberry Pi.
- **ForzaETH ROS2 Autonomous Driving** Oct. 2025 – Present
Team Project
– Developed a LiDAR-based Gap Reference Pursuit (GRP) obstacle-avoidance and dynamic-speed system using ROS2 perception & control loops.
- **Computer Vision Mini Projects** Oct. 2025 – Nov. 2025
Individual Project
– Implemented Korean coin detection and playing-card recognition using classical CV techniques.
– Coin detection using adaptive thresholding, morphology, and contour extraction.
– Card rank/suit recognition for overlapping cards using grayscale, Gaussian blur, adaptive thresholding, and template matching.
- **Nighttime Tourism Data Analysis Competition** Jun. 2025 – Jul. 2025
Team Project
– Proposed strategies for Jeollabuk-do based on multi-source tourism datasets(travel surveys, tourism time-stay logs, visitor statistics).
– Applied correlation, regression, KDE spatial density, and time-series trend analysis.
– Developed tailored strategies for Gimje, Sunchang, and Imsil.
- **Crime Prevention Infrastructure Impact Analysis** May 2025 – Jun. 2025
Team Project
– Analyzed the impact of crime-prevention infrastructure on crime incidence.
– Conducted correlation analysis, regression modeling, and visual analytics.

- **“Babedeuk” Restaurant Information Web App (UMC)** *Apr. 2024 – Aug. 2024*
UMC(University MakeUs Challenge) – Team Project
 - Developed a restaurant information web app using React and OAuth-based social login.
 - Implemented detail pages and collaborated via Git-based workflows.
- **Fashion MNIST Classification Competition** *Nov. 2023 – Dec. 2023*
Team Project
 - Improved accuracy via SVM, PCA variance experiments, and HOG feature extraction.

STUDY ACTIVITIES

- **OPIc Study** *Sep. 2025 – Present*
 - Twice-weekly speaking practice to improve fluency, spontaneous responses, and topic expansion.
- **UMC Frontend Study** *Mar. 2024 – Jun. 2024*
 - Studied full frontend development using React
 - Covered component routing, state management, API handling, and Debouncing/Throttling, storage, Responsive Web Design, Redux.
- **Deep Learning with PyTorch** *Mar. 2024 – Jun. 2024*
 - Independently studied tensors, linear/logistic regression, softmax regression, perceptrons, backpropagation, vanishing gradients, RNN, and LSTM.
- **HTML/CSS Course Study** *Mar. 2024 – Apr. 2024*
 - Completed Inflearn’s foundational HTML/CSS course
- **Machine Learning Study** *Jan. 2024 – Apr. 2024*
 - Weekly study sessions based on “Python Machine Learning Guide”
 - Covered NumPy, Pandas, scikit-learn, cross-validation, preprocessing, evaluation metrics, Decision Trees, ensemble models, and hyperparameter tuning.
- **Baekjoon Algorithm Study** *Jan. 2024 – Mar. 2024*
 - Solved one coding problem per day

TECHNICAL SKILLS AND INTERESTS

Areas of Interest: Multilingual NLP, LLM Agents, Reasoning, Self-improving AI Systems, RAG
Languages: C++, C, Python, JavaScript, HTML/CSS
Developer Tools: Git, ROS2, Raspberry Pi, Jupyter, VS Code
Frameworks: PyTorch, HuggingFace Transformers, React
Coursework: Advanced AI Mathematics, Machine Learning, Data Science, Advanced Neural Networks, NLP, LLM, Computer Vision

SCHOLARSHIPS & AWARDS

- **Baekma Academic Excellence Scholarship** – Top 25% performance *2023 Fall*
- **Grand Prize** – Hyundai Mobis Autonomous Driving Competition *2020*